Machine Learning Identification of the Phase Transition and Order Parameter in the 2D Ising Model

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April 2026
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1. Introduction

Phase transitions are among the most important phenomena in condensed matter and statistical physics because they show how simple microscopic interactions can produce dramatic macroscopic behavior. A classic example is the ferromagnetic phase transition, where a system changes from a disordered high-temperature phase to an ordered low-temperature phase with nonzero magnetization. Near the critical point, fluctuations become large and correlations extend across many length scales, making the transition especially interesting from both a physical and computational perspective.

A standard model for studying this behavior is the two-dimensional ferromagnetic Ising model, in which each lattice site carries a spin s_i=±1. For nearest-neighbor interactions on a square lattice, the Hamiltonian is

$$H = -J \sum_{<i,j>} s_i s_j$$

where J > 0 favors alignment of neighboring spins. At high temperature, thermal fluctuations dominate and the spins appear disordered, corresponding to a paramagnetic phase. At low temperature, aligned domains emerge and the system develops a nonzero magnetization. The natural order parameter is the magnetization,

$$m = \frac{1}{N} \Sigma_i s_i$$

and for the two-dimensional square-lattice Ising model the exact critical temperature is

$$T_c = \frac{2}{ln(1+\sqrt{2})} \approx 2.269$$

These properties make the Ising model an ideal benchmark for machine-learning studies of phase transitions because both the critical temperature and the physical order parameter are known exactly.

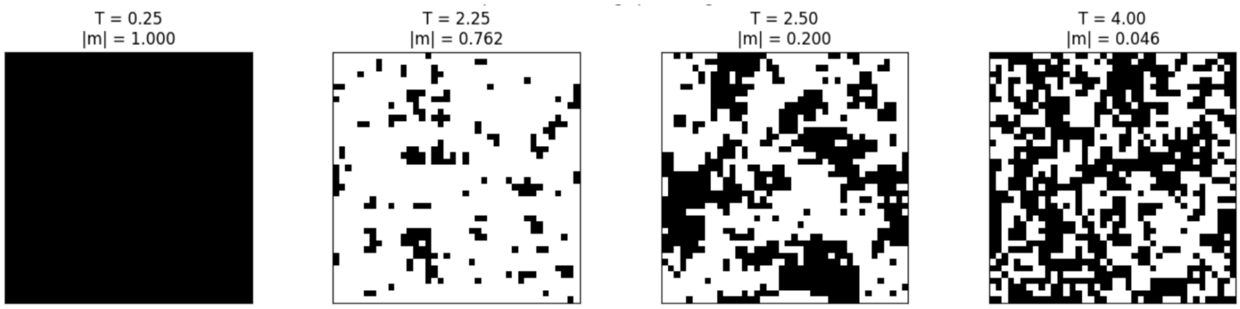

Figure 1: Representative 40×40 Ising spin configurations at selected temperatures. At low temperature the spins are strongly aligned, indicating the ordered ferromagnetic phase. Near the critical region, large fluctuating domains appear, while at high temperature the configurations become disordered, consistent with the paramagnetic phase.

In recent years, machine learning has become an increasingly useful tool for identifying phases of matter and detecting phase transitions directly from raw many-body configurations. Unsupervised approaches are especially appealing because they do not require prior labeling of phases and can instead extract dominant patterns directly from the data. Wang (2016) showed that principal component analysis, or PCA, applied to raw Ising configurations can recover physically meaningful low-dimensional features and identify the phase transition, with the leading principal component corresponding to the magnetization mode. Hu, Singh, and Scalettar (2017) further emphasized that PCA can identify transition regions and symmetry-breaking patterns while also showing that its success depends on the kind of configurations supplied to the algorithm and on whether the relevant physics is accessible through linear features. Wetzel (2017) likewise found that PCA and variational autoencoders are especially promising unsupervised tools because their learned latent variables can track the correct order parameter and reveal phase transitions without prior labeling.

In this project, a provided Monte Carlo dataset of 40×40 Ising spin configurations sampled across a range of temperatures was analyzed using both conventional observables and machine-learning methods. The first goal was to determine the critical region and estimate the critical temperature from the data. The second goal was to investigate whether an unsupervised method could recover the physically meaningful order parameter directly from raw configurations. To address these questions, the study first examined representative configurations and standard physical observables such as the absolute magnetization and a connected susceptibility. A logistic regression classifier was then trained on clearly ordered and clearly disordered states and applied across all temperatures to identify the phase boundary. Finally, PCA was used to test whether the dominant structure in the data corresponds to the magnetization mode.

Overall, the 2D Ising model provides a controlled setting for understanding how machine learning can be applied to many-body physics. By comparing conventional observables with supervised and unsupervised learning results, this study aims to show both how machine learning can detect a phase transition and how it can recover the underlying order-parameter structure from the data itself. This broader motivation is important because, in more complicated systems, the order parameter may be less obvious or not known in advance.

2. Methods

The goal of this project was to study the phase transition of the two-dimensional ferromagnetic Ising model using both conventional physical observables and machine-learning methods. The analysis was designed to address two related questions: first, whether the transition region and critical temperature could be detected from the spin configurations, and second, whether an unsupervised learning method could recover the physically meaningful order parameter from the raw data. In keeping with the project guidelines, the methods were chosen to be simple, interpretable, and reproducible, while still incorporating standard machine-learning practices such as train-test splitting, cross-validation, and hyperparameter tuning.

The provided dataset consisted of Monte Carlo-sampled spin configurations for a 40×40 square lattice at multiple temperatures. Each configuration was first unpacked into explicit spin values $s_i = \pm1$, and the corresponding temperature was assigned based on the ordering of the dataset. Since each lattice contains $N=1600$ spins, every sample was represented as a vector in a 1600-dimensional configuration space. This representation is consistent with the literature, where raw spin configurations are treated as high-dimensional input data for feature extraction and clustering or classification methods.

Before applying machine learning, standard physical observables were computed as a baseline. For each configuration, the magnetization was calculated, and temperature-dependent averages of the absolute magnetization were used to visualize the transition from the ordered ferromagnetic phase to the disordered paramagnetic phase. A connected susceptibility-like quantity was also evaluated in order to characterize fluctuations near the transition. This step served two purposes: it verified that the phase transition was visible in the data using conventional statistical-physics quantities, and it provided a benchmark against which the machine-learning results could later be compared. The literature on PCA for the Ising model similarly emphasizes that quantified leading components can be compared directly with magnetization- and susceptibility-like behavior.

$$\chi_{|m|} = \frac{N}{T}\left(\langle m^2 \rangle - \langle |m| \rangle^2\right)$$

To estimate the phase boundary using supervised learning, a logistic regression classifier was trained to distinguish clearly ordered and clearly disordered states. Only configurations at sufficiently low temperatures and sufficiently high temperatures were included in the supervised training set, while intermediate temperatures near the critical region were excluded from training and reserved for later probing. This choice ensured that the classifier learned a clean phase distinction rather than ambiguous near-critical examples. Because the Ising model has a global spin-flip symmetry, configurations were first aligned so that their total magnetization was nonnegative. This preprocessing step removes the trivial Z_2 ambiguity between positively and negatively magnetized ordered states and makes the classification problem more physically meaningful.

The supervised dataset was divided into training and test sets, and the classifier hyperparameters were tuned using cross-validation on a smaller subset of the training data. In the final implementation, stochastic logistic regression was used for computational efficiency while preserving the interpretation of a linear classifier. Model performance was evaluated using test accuracy, a classification report, and a confusion matrix. After training, the classifier was applied to all configurations in the dataset, and the mean predicted probability of belonging to the ordered phase was computed as a function of temperature. The critical temperature was then estimated from the point where this mean probability crossed 0.5. This follows the general strategy discussed in the machine-learning literature, where supervised or unsupervised low-dimensional indicators are used to identify phase boundaries and locate transition regions.

To identify the order-parameter structure without using phase labels, principal component analysis (PCA) was applied to a balanced subset of the spin configurations spanning all temperatures. PCA is a linear dimensional-reduction method that finds orthogonal directions of maximal variance in the data. In the context of spin systems, each configuration is treated as a point in a high-dimensional space, and PCA determines the directions along which the configurations vary most strongly. The projection of the data onto these directions yields principal component scores, which can be examined as possible indicators of phase behavior. As emphasized in prior work, PCA is especially effective when the dominant physics is captured by a linear global mode, as in the ferromagnetic Ising model where the leading component is expected to correspond to magnetization.

$$X^T X w_l = \lambda_l w_l$$

$$y_l = X w_l$$

Here, X denotes the centered data matrix, w_l is the ℓ-th principal component weight vector, λ_l is the corresponding eigenvalue, and y_l is the projection of the configurations onto that component. In practice, the explained variance ratios of the leading components were computed, the first principal component was reshaped back onto the lattice for visualization, and its scores were compared directly with the magnetization. The temperature dependence of the mean absolute first principal component was also examined to determine whether it behaves like an order parameter. This procedure closely follows the interpretation used in the Ising-model literature, where the leading PCA component is found to be nearly uniform across the lattice and strongly associated with the global magnetization mode.

Finally, the results from conventional observables, supervised classification, and PCA were compared through plots and summary tables. This allowed the transition region identified by machine learning to be assessed against known statistical-physics expectations and made it possible to evaluate how well the unsupervised representation captured the order parameter of the system.

3. Results

3.1 Data Exploration and Physical Observables

Before applying any machine-learning method, the dataset was examined using standard physical observables to check that the phase transition is already visible in the raw spin configurations. Figure 1 gives a qualitative picture of this behavior: at low temperature the system is strongly ordered, near the critical region large fluctuating domains appear, and at high temperature the configurations become disordered.

To quantify this change, the absolute value of the magnetization was averaged at each temperature. As expected for the ferromagnetic Ising model, the average absolute magnetization stays close to 1 at low temperature and then drops sharply in the transition region. This is consistent with the loss of long-range ferromagnetic order as the temperature increases toward and above the critical point. Since the exact critical temperature of the two-dimensional square-lattice Ising model is about 2.269, the sharp drop in the average absolute magnetization is a good first sign that the dataset captures the expected transition correctly.

In addition to magnetization, a connected susceptibility-like quantity was computed to better describe the fluctuations near the transition. This modified form is more suitable for the present dataset because the ordered configurations include both positive- and negative-magnetization sectors. The resulting curve shows a clear peak in the critical region, which is exactly what would be expected near a phase transition. In this dataset, the peak appears at about temperature 2.25, which is very close to the exact Onsager value.

Overall, these plots show that the dataset captures the expected behavior of the two-dimensional Ising model. The phase transition is already visible from simple physical observables before any machine-learning model is used. This gives a strong baseline for the rest of the report and makes it easier to compare the machine-learning results with standard statistical-physics expectations.

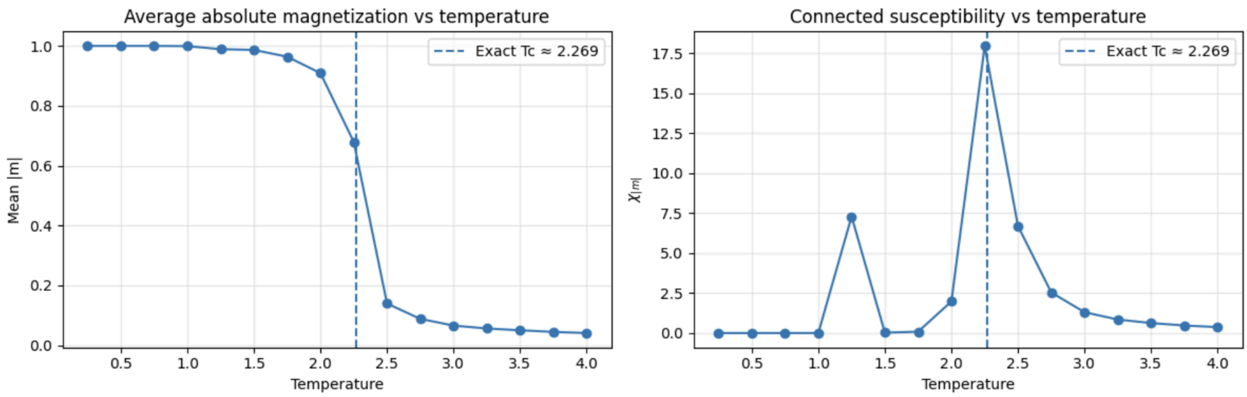

Figure 2: Temperature dependence of the average absolute magnetization and connected susceptibility for the 40 by 40 Ising dataset. The magnetization decreases sharply across the transition region, while the connected susceptibility develops a peak near temperature 2.25, close to the exact critical temperature of about 2.269.

3.2 Supervised Learning: Logistic Regression

To detect the phase boundary using supervised learning, a logistic regression classifier was trained on configurations that were clearly far from the critical region. Low-temperature configurations were labeled as ordered, while high-temperature configurations were labeled as disordered. Configurations near the transition were left out of the training set so that the model would learn the difference between the two phases from clean examples rather than from strongly fluctuating intermediate cases. Because the ferromagnetic Ising model has a global spin-flip symmetry, the configurations were first aligned so that their total magnetization was nonnegative. This removes the trivial difference between positively and negatively magnetized ordered states and makes the classification problem more physically meaningful.

The supervised model performed extremely well on the held-out test set. The cross-validation score was about 0.9995, and the final test accuracy was about 0.9993. The confusion matrix shows that almost all configurations were assigned to the correct phase, with only a very small number of misclassifications. This means that once the global spin-flip ambiguity is removed, the ordered and disordered phases are easily separable by a simple linear classifier when the training data come from temperatures well away from the critical region.

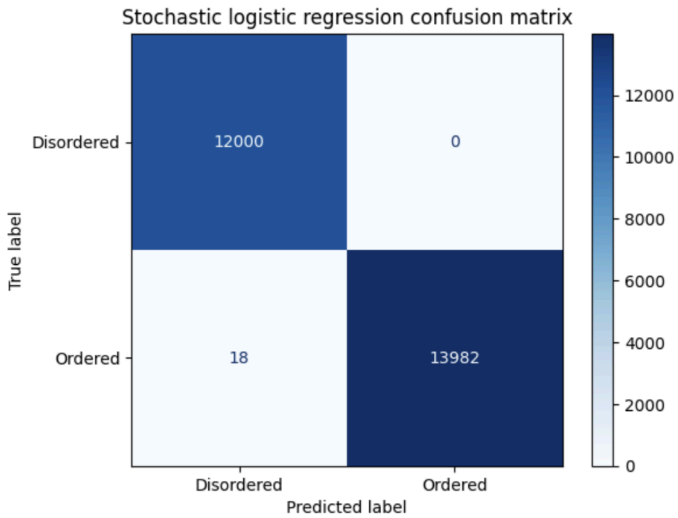

Figure 3: Confusion matrix for the stochastic logistic regression classifier. The classifier separates the ordered and disordered phases with very high accuracy when trained on configurations far from the critical region.

However, classification accuracy alone does not tell us the critical temperature. To estimate the transition region, the trained classifier was applied to the full dataset across all temperatures, and the average predicted probability that a configuration belongs to the ordered phase was computed at each temperature. The resulting curve drops sharply from values close to 1 at low temperature to values close to 0 at high temperature. This shows that the classifier is picking up the phase change in the configurations, not just fitting the training labels.

Using linear interpolation at the point where the probability curve crosses 0.5, the machine-learning estimate of the critical temperature was found to be about 2.10. This is lower than the exact value of about 2.269, but it still lies in the correct transition region. The difference is most likely due to the coarse temperature spacing in the dataset and the fact that the model was trained only on temperatures far from the critical region. Even so, the sharp change in the classifier output shows that supervised learning successfully detects the phase boundary in the Ising model.

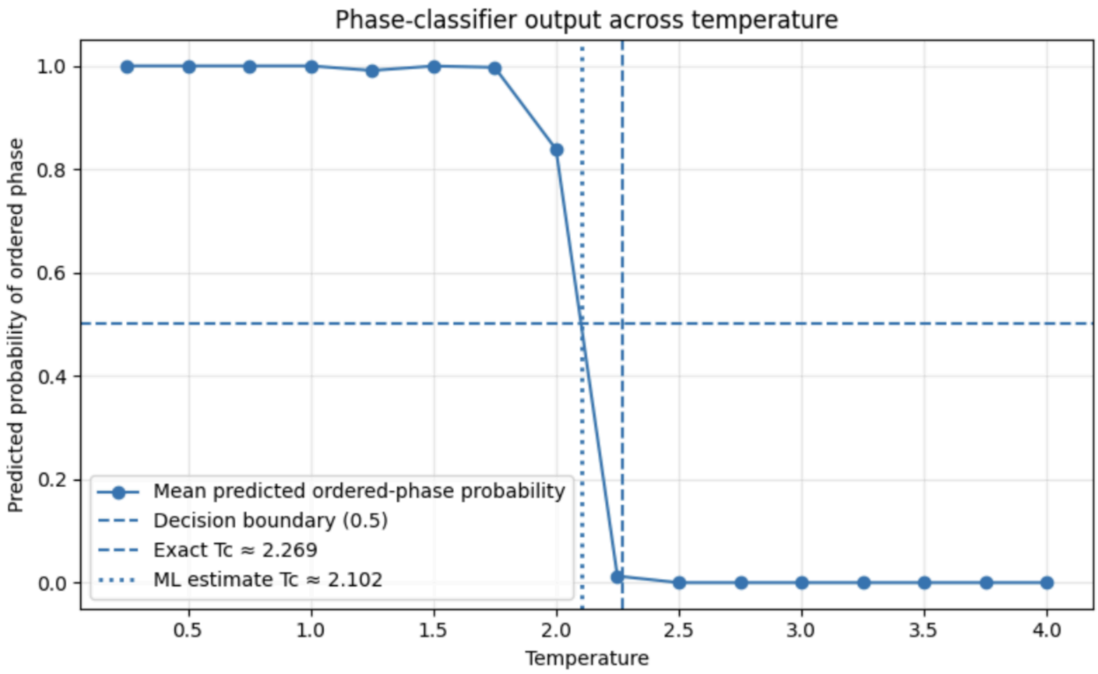

Figure 4: Average predicted probability of the ordered phase as a function of temperature. The classifier output changes sharply across the transition region. Linear interpolation of the 0.5 crossing gives a machine-learning estimate of the critical temperature of approximately 2.10.

These supervised-learning results show that the phase boundary can be detected directly from raw spin configurations using a simple and interpretable linear model. At the same time, the difference between the machine-learning estimate and the exact critical temperature makes it clear that high classification accuracy does not automatically mean the transition point will be estimated perfectly. For that reason, it is useful to compare the supervised results with both the conventional observables and the PCA analysis in the next subsection.

3.3 Unsupervised Learning: PCA and the Order Parameter

The second major goal of this project was to test whether an unsupervised learning method could recover the physically meaningful order parameter directly from raw spin configurations. For this purpose, principal component analysis, or PCA, was applied to a balanced subset of configurations spanning all temperatures. Unlike supervised learning, PCA does not use phase labels. Instead, it identifies the directions of greatest variance in the data and projects the configurations onto those directions. This makes it a natural method for checking whether the dominant structure of the Ising dataset is related to magnetization.

The PCA results were very clear. The first principal component explained about 51.7 percent of the total variance, while the second and third components contributed only a very small amount by comparison. This means that the dataset is dominated by one main large-scale pattern rather than many competing ones. When the first principal component was reshaped back onto the 40 by 40 lattice, its weight pattern was found to be nearly uniform across the whole system. That is exactly what would be expected if PCA had identified a global magnetization-like mode.

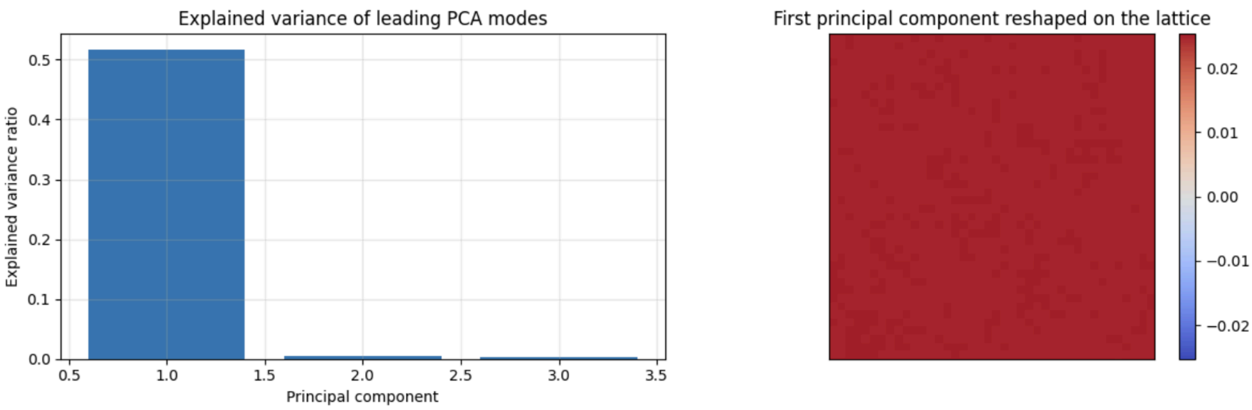

Figure 5: Explained variance of the leading principal components and the first principal component reshaped onto the lattice. The first component explains a large fraction of the total variance, and its nearly uniform spatial structure indicates that it corresponds to a global magnetization-like mode.

To test this more directly, the first principal component scores were compared with the magnetization of the same configurations. The correlation between the first principal component and the magnetization was essentially 1, showing an almost perfect linear relationship. This is one of the strongest results in the project because it shows that PCA, without using any labels, recovers the same physical quantity that is already known to be the order parameter of the ferromagnetic Ising model.

The temperature dependence of the mean absolute value of the first principal component gives even more support for this interpretation. As temperature increases, the magnitude of the first principal component stays large at low temperature and then decreases sharply across the transition region, closely following the behavior of the average absolute magnetization. The scatter plot of the first principal component against magnetization also shows an almost perfect straight-line relationship. Together, these results show that PCA is not just giving an abstract reduced representation of the data. It is extracting the physically relevant order-parameter structure.

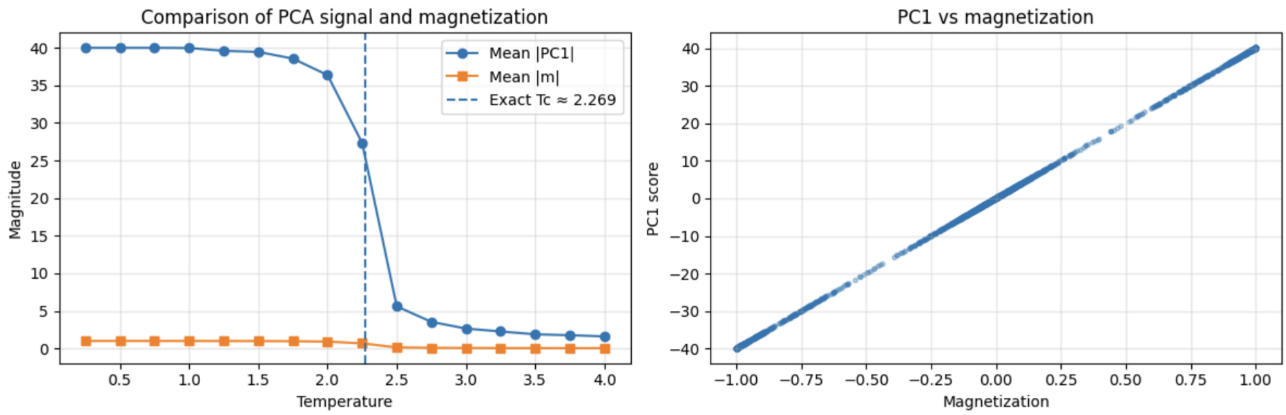

Figure 6: Comparison of the PCA signal and magnetization. The left panel shows that the mean absolute value of the first principal component decreases across the transition in the same way as the average absolute magnetization. The right panel shows an almost perfect linear relationship between the first principal component and magnetization.

The clearest unsupervised result in the report comes from PCA. While the supervised classifier successfully detected the phase boundary, PCA went a step further by showing what physical quantity controls the dominant structure in the dataset. In this case, that quantity is magnetization. This means that, for the two-dimensional ferromagnetic Ising model, unsupervised learning can recover the correct order parameter directly from raw configurations without using any labels. More broadly, this suggests that similar methods could be useful in systems where the order parameter is less obvious from the start.

3.4 Summary of Main Quantitative Results

To compare the conventional observables and machine-learning results more directly, the main numerical outcomes of the project were collected into a single summary table. This table includes the exact critical temperature, the transition estimate from the connected susceptibility, the machine-learning estimate from the logistic regression transition curve, the classifier accuracy values, and the PCA results related to the order parameter.

The summary table confirms the overall picture from the earlier subsections. The connected susceptibility gives a transition estimate near temperature 2.25, which is close to the exact critical temperature of about 2.269. The supervised classifier achieves extremely high cross-validation and test accuracy when separating clearly ordered and clearly disordered states. When applied across all temperatures, it also produces a sharp phase boundary, although its interpolated estimate of the critical temperature is somewhat lower than the exact value.

The strongest result in the table is still the unsupervised-learning outcome. The first principal component explains about 51.7 percent of the variance, and its correlation with magnetization is essentially 1. This means that principal component analysis recovered the same physical quantity that is already known to be the order parameter of the ferromagnetic Ising model.

The summary table brings the main results together in one place. The connected susceptibility gives a transition estimate close to the exact value, the supervised classifier identifies the transition region very clearly, and the PCA analysis provides the strongest direct physical interpretation. Overall, the results show that conventional observables, supervised learning, and unsupervised learning all identify the phase transition, but PCA gives the clearest link to the underlying order-parameter structure.

Quantity	Value
Exact critical temperature	2.269000
Susceptibility-based Tc estimate	2.250000
ML classifier Tc estimate	2.102493
Absolute error of ML Tc	0.166507
Best CV accuracy (logistic regression)	0.999500
Test accuracy (logistic regression)	0.999308
Correlation between PC1 and magnetization	1.000000
Explained variance ratio of PC1	0.517029

Table 1: Summary of the main quantitative results of the project, including the exact critical temperature, susceptibility-based and machine-learning-based transition estimates, classifier performance, and principal component analysis results.

4. Discussion and Summary

This project examined the phase transition of the two-dimensional ferromagnetic Ising model using both standard physical observables and machine-learning methods. Overall, all three approaches used here - conventional statistical-physics quantities, supervised learning, and unsupervised learning, identified the transition region in a meaningful way. The absolute magnetization and connected susceptibility gave a clear physics baseline, with the susceptibility peak appearing near temperature 2.25, close to the exact critical temperature of about 2.269. This confirms that the supplied Monte Carlo dataset captures the expected behavior of the Ising model.

The supervised-learning results showed that a simple logistic regression classifier can separate clearly ordered and clearly disordered configurations with extremely high accuracy. After aligning the spin configurations to remove the global spin-flip ambiguity, the classifier achieved near-perfect cross-validation and test performance. When the trained model was applied across all temperatures, it produced a sharp transition curve in the predicted probability of the ordered phase. This shows that the classifier learned a meaningful phase distinction directly from the raw spin configurations. At the same time, the machine-learning estimate of the critical temperature, about 2.10, was lower than the exact value. This suggests that very high classification accuracy does not automatically guarantee a precise estimate of the critical point. In this case, the difference is likely related to the coarse temperature spacing of the dataset and the fact that the model was trained only on temperatures far from the critical region.

The strongest result of the project came from the unsupervised analysis. Principal component analysis revealed one dominant component that explained about 51.7 percent of the variance, while the remaining leading components were much smaller. The first principal component was nearly uniform across the lattice and was almost perfectly correlated with magnetization. Its temperature dependence also closely followed the behavior of the average absolute magnetization. This shows that PCA recovered the physically meaningful order parameter directly from raw configurations without using phase labels.

Overall, two main conclusions stand out from the results. First, machine learning can successfully detect the phase boundary in the two-dimensional Ising model. Second, unsupervised learning can recover the underlying order-parameter structure directly from the data. The comparison between the supervised and unsupervised methods is especially useful here. The supervised classifier was very effective once it was trained on clean examples, but its estimate of the critical temperature still depended on how the transition curve was inferred. PCA, on the other hand, gave a more direct physical interpretation because its dominant mode matched the magnetization almost exactly. That is an important reminder that machine-learning results are most useful when they can also be connected back to interpretable physics.

There are several natural ways this work could be extended. One improvement would be to use a finer temperature grid near the critical region in order to get a more accurate machine-learning estimate of the critical temperature. Another would be to test nonlinear unsupervised methods such as autoencoders or variational autoencoders, which have been shown to capture phase-transition structure in cases where linear PCA may be less effective. More generally, the same methods could be applied to systems with more complicated symmetry breaking or less obvious order parameters, where unsupervised learning may provide genuinely new physical insight.

In summary, the two-dimensional Ising model turned out to be a very useful benchmark for studying machine learning in many-body physics. Conventional observables identified the expected transition, supervised learning detected a sharp phase boundary, and principal component analysis recovered the known order parameter directly from the spin configurations. Together, these results show that machine learning can do more than just classify phases, it can also help reveal the physical structure that drives a phase transition.

5. References

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